

**Trig 5****Review**

1.

$$\begin{aligned}\frac{2}{3\sqrt{2}} &= 2 \frac{3\sqrt{2}}{(3\sqrt{2})^2} \\ &= \frac{6\sqrt{2}}{18} = \frac{1}{3}\sqrt{2}\end{aligned}$$

2.

$$\begin{aligned}\int_{-1}^{-2} \left(1 - \frac{1}{x^3}\right) dx &= \int_{-1}^{-2} (1 - x^{-3}) dx \\ &= \left[ x + \frac{1}{2x^2} \right]_{-1}^{-2} \\ &= \left( -2 + \frac{1}{2(-2)^2} \right) - \left( -1 + \frac{1}{2(-1)^2} \right) \\ &= \frac{11}{8}\end{aligned}$$

3. When  $g(4x)$  crosses the coordinate axes,  $4x = -2$  or  $g(4x) = -1$ .So  $g(4x)$  crosses x-axis at  $x = -\frac{1}{2}$  (point  $(-\frac{1}{2}, 0)$ ).

And it still crosses the y-axis at 0, -1 because the y-intercept does not change. The graph is squished along the x-axis.

4.  $f(x) = x^{-1} - x^{-2} \Rightarrow f'(x) = -x^{-2} + 2x^{-3}$ .When  $y' = 0$ , the gradient is 0.

$$\begin{aligned}\frac{2}{x^3} - \frac{1}{x^2} &= 0 \\ \frac{x^3}{2} - x^2 &= 0 \\ x^2(x - 2) &= 0\end{aligned}$$

But  $f(0)$  is undefined, so  $x = 2$  when the gradient is 0.

5.

$$\begin{aligned}\cos \theta &= 0.1 \\ \theta &= 0.995^c, 2.15^c\end{aligned}$$

**Consolidation**

1. i.

$$\begin{aligned}\cos 2x &= 2 \cos^2 x - 1 \\ &= 0.72 - 1 \\ &= -0.28\end{aligned}$$

ii.

$$\begin{aligned}\cos 2x &= \cos^2 x - \sin^2 x \\ \sin x &= \sqrt{(\cos x)^2 - \cos 2x} \\ &= \sqrt{(0.6)^2 + 0.28} \\ &= 0.8\end{aligned}$$

$$\begin{aligned}
 \sin 2x &= 2 \cos x \sin x \\
 &= 2(0.6)(0.8) \\
 &= 0.96
 \end{aligned}$$

$$\begin{aligned}
 \sin 3x &= \sin(2x + x) \\
 &= \sin(2x) \cos(x) + \cos(2x) \sin(x) \\
 &= 2 \sin(x)(1 - \sin^2 x) + (1 - 2 \sin^2 x) \sin(x) \\
 &= 2 \sin x - 2 \sin^3 x + \sin x - 2 \sin^3 x \\
 &= 3 \sin x - 2 \sin^3 x \\
 &= 3 \sin x - 2(\sin x)^3 \\
 &= 3 \times 0.8 - 2 \times 0.512 \\
 &= 2.912
 \end{aligned}$$

$$\begin{aligned}
 2. \quad \sin(x + y) &= \sin x \cos y + \cos x \sin y \\
 &= \sin\left(\arccos\left(\frac{3}{5}\right)\right) \times \cos\left(\arcsin\left(\frac{12}{13}\right)\right) + \frac{12}{13} \times \frac{3}{5} \\
 &= \left(\frac{56}{65}\right)^c
 \end{aligned}$$

$$\begin{aligned}
 \cos(x - y) &= \cos x \cos y - \sin x \sin y \\
 &= \frac{3}{5} \times \cos\left(\arcsin\left(\frac{12}{13}\right)\right) + \sin\left(\arccos\left(\frac{3}{5}\right)\right) \times \frac{12}{13} \\
 &= \left(\frac{63}{65}\right)^c
 \end{aligned}$$

$$\begin{aligned}
 \tan 2x &= \frac{\sin 2x}{\cos 2x} \\
 &= \frac{2 \sin x \cos x}{\cos^2 x - \sin^2 x} \\
 &= \frac{2(\cos(\arcsin(\frac{12}{13})))(\frac{3}{5})}{(\frac{3}{5})^2 - \cos(\arcsin(\frac{12}{13}))^2} \\
 &= \left(\frac{975}{448}\right)^c
 \end{aligned}$$

$$3. \quad \text{i. } \cos 5x = 0.5$$

$$\begin{aligned}
 5x &= \frac{1}{3}\pi, \frac{5}{3}\pi, \frac{7}{3}\pi, \frac{11}{3}\pi, \frac{13}{3}\pi, \frac{17}{3}\pi, \frac{19}{3}\pi, \frac{23}{3}\pi, \frac{25}{3}\pi, \frac{29}{3}\pi \\
 x &= \frac{1}{15}\pi, \frac{1}{3}\pi, \frac{7}{15}\pi, \frac{11}{15}\pi, \frac{13}{15}\pi, \frac{17}{15}\pi, \frac{19}{15}\pi, \frac{23}{15}\pi, \frac{5}{3}\pi, \frac{29}{15}\pi
 \end{aligned}$$

$$\text{ii. } \sin 6x = 1$$

$$\begin{aligned}
 6x &= \frac{\pi}{2}, \frac{5\pi}{2}, \frac{9\pi}{2}, \frac{13\pi}{2}, \frac{17\pi}{2}, \frac{21\pi}{2} \\
 x &= \frac{\pi}{12}, \frac{5\pi}{12}, \frac{3\pi}{4}, \frac{13\pi}{12}, \frac{17\pi}{12}, \frac{21\pi}{12}
 \end{aligned}$$

$$\begin{aligned}
4. \quad &= \cos(A - (A - B)) \\
&= \cos B
\end{aligned}$$

$$\begin{aligned}
5. \quad \text{i.} \quad &1 - 2 \sin x - 4(1 - 2 \sin^2 x) = 0 \\
&8 \sin^2 x - 2 \sin x - 3 = 0 \\
&\sin x = \frac{3}{4} \text{ or } \sin x = -\frac{1}{2} \\
&x = 30^\circ, 48.6^\circ, 131^\circ, 150^\circ
\end{aligned}$$

$$\begin{aligned}
\text{ii.} \quad &\sin 2x - \frac{\sin x}{\cos x} = 0 \\
&\sin 2x \cos x - \sin x = 0 \\
&2 \sin x \cos^2 x - \sin x = 0 \\
&2 \sin x(1 - \sin^2 x) - \sin x = 0 \\
&2 \sin x - 2 \sin^3 x - \sin x = 0 \\
&\sin x - 2 \sin^3 x = 0 \sin x(1 - 2 \sin^2 x) = 0
\end{aligned}$$

Therefore

$$\begin{aligned}
&\sin x = 0 \\
&x = 0^\circ, 180^\circ
\end{aligned}$$

or

$$\begin{aligned}
&\sin x = \pm \frac{1}{\sqrt{2}} \\
&x = 45, 135, 225, 315
\end{aligned}$$

So  $x = 0, 45, 135, 180, 225, 315$

$$\begin{aligned}
\text{iii.} \quad &\tan 2x + \tan x = 0 \\
&\frac{2 \tan x}{1 - \tan^2 x} + \tan x = 0 \\
&\frac{2 \tan x \times \tan x(1 - \tan^2 x)}{1 - \tan^2 x} = 0 \\
&\frac{2 \tan^2 x - 2 \tan^4 x}{1 - \tan^2 x} = 0 \\
&\tan^2 x(2 - 2 \tan^2 x) = 0
\end{aligned}$$

So  $\tan x = 0$  or  $\tan x = 1$  or  $\tan x = -1$ .

And  $x = 0, 45, 135, 180, 225, 315, 360$ .

$$\begin{aligned}
6. \quad \text{i.} \quad &\cos^4 x - \sin^4 x = (\cos^2 x - \sin^2 x)(\cos^2 x + \sin^2 x) \\
&= (\cos 2x)(1) \\
&= \cos 2x
\end{aligned}$$

$$\begin{aligned}
 \text{ii.} \quad \frac{\cos x - \sin x}{\cos x + \sin x} &= (\cos x - \sin x) \frac{\cos x + \sin x}{(\cos x + \sin x)^2} \\
 &= \frac{\cos^2 x - \sin^2 x}{\cos^2 x + \sin^2 x + 2 \cos x \sin x} \\
 &= \frac{\cos 2x}{1 + \sin 2x}
 \end{aligned}$$

$$\begin{aligned}
 \text{iii.} \quad \frac{\sin x}{1 + \cos x} + \frac{1 - \cos x}{\sin x} &= \frac{2 \sin \frac{x}{2} \cos \frac{x}{2}}{1 + 2 \cos^2 \frac{x}{2} - 1} + \frac{1 - (1 - 2 \sin^2 \frac{x}{2})}{2 \sin \frac{x}{2} \cos \frac{x}{2}} \\
 &= \tan \frac{x}{2} + \tan \frac{x}{2} \\
 &= 2 \tan \frac{x}{2}
 \end{aligned}$$

$$\begin{aligned}
 7. \quad \sin 3x &= \sin 2x \cos x + \sin x \cos 2x \\
 &= 3 \sin x - 4 \sin^3 x = \sin x
 \end{aligned}$$

So

$$4 \sin^3 x - 2 \sin x = 0 \implies 2 \sin x (\sin^2 x - 1) = 0$$

So  $\sin x = 0$  or  $\sin x = 1$  or  $\sin x = -1$ . And  $x = 0^\circ, 90^\circ, 180^\circ$ .  $\sin x = -1$  has no solutions between  $0^\circ$  and  $180^\circ$ .

8. i.  $\sin 2\theta = 2\theta$   
 ii.  $\tan 3\theta = 3\theta$   
 iii.  $\sin \theta \cos \theta = \theta - \frac{1}{2}\theta^3$   
 iv.  $\tan \theta \cos 2\theta = \theta - 2\theta^3$   
 v.  $1 - \cos 2\theta = 2\theta^2$

### Exam Questions

$$\begin{aligned}
 1. \quad 1 + \cos \theta - 3 \cos^2 \theta &= 1 + \left(1 - \frac{1}{2}\theta^2\right) - 3\left(1 - \frac{1}{2}\theta^2\right)^2 \\
 &= 2 - \frac{1}{2}\theta^2 - 3\left(1 - \theta^2 + \frac{\theta^4}{4}\right) \\
 &= -1 + \frac{5}{2}\theta^2 - \frac{3}{4}\theta^4
 \end{aligned}$$

Since  $\theta$  is a small angle,  $\theta^4 \approx 0$ . Therefore

$$1 + \cos \theta - 3 \cos^2 \theta \approx -1 + \frac{5}{2}\theta^2$$

2. a.

$$(\cos 2x)^2 = 0.25$$

$$\cos 2x \in \{-0.5, 0.5\}$$

$$1 - 2 \sin^2 x \in \{-0.5, 0.5\}$$

$$2 \sin^2 x \in \{0.5, 1.5\}$$

$$\sin^2 x \in \{0.25, 0.75\}$$

$$\sin x \in \left\{ 0.5, \frac{\sqrt{3}}{2} \right\}$$

$$x \in \{30, 60, 120, 150\}$$

$$\begin{aligned} \text{b. i. } \frac{\cos \theta}{\cos \theta - \sin \theta} - \frac{\cos \theta}{\cos \theta + \sin \theta} &= \frac{(\cos \theta)(\cos \theta + \sin \theta) - (\cos \theta)(\cos \theta - \sin \theta)}{\cos^2 \theta - \sin^2 \theta} \\ &= \frac{\cos^2 \theta + \sin \theta \cos \theta - \cos^2 \theta + \sin \theta \cos \theta}{\cos 2\theta} \\ &= \frac{2 \sin \theta \cos \theta}{\cos 2\theta} \\ &= \frac{\sin 2\theta}{\cos 2\theta} \\ &= \tan 2\theta \end{aligned}$$

ii.

$$\tan 2\theta = 1$$

$$2\theta = 45^\circ, 225^\circ, 405^\circ, 585^\circ$$

$$\theta = 22.5^\circ, 112.5^\circ, 202.5^\circ, 292.5^\circ$$